

3 Chapter 3: Time-Domain Analysis of Discrete-Time Systems

3.1 Introduction

Discrete-time signals, unlike continuous-time signals, use a discretized domain n rather than a continuous domain t , with a range specified by $x[n]$, which may be continuous or discrete itself. Any sequence of numbers with a discrete domain can be considered a discrete-time signal, and though discrete-time signals and their corresponding discrete-time systems have similar general properties to the continuous-time systems we analyzed in Chapter 2, there are a number of important differences that we'll learn about in this chapter.

It's worth noting that discrete-time signals often arise as a result of *sampling* continuous-time signals, which effectively converts them to discrete-time signals. Uniform sampling is the ideal, and the only process we will be working with in this class, meaning that the continuous-time signal will be sampled at a consistent rate of T , such that $x[n] = x(nT)$.

Sampling and *quantization* are different processes. Sampling affects the domain and picks out periodic values of a continuous signal, converting it to a discrete-time signal. Quantization affects the range, and converts an analog signal to a digital one by reducing its real value to the closest available digital (pick from a finite array of options) value.

For discrete-time signals, the measurement of energy and power are almost identical to that for continuous time signals, but using sums instead of integrals:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$
$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2.$$

Similar to continuous-time signals, periodic discrete-time signals may have their power calculated over a single period N_0 rather than taking the limit as $N \rightarrow \infty$. For discrete-time signals, we may define a *fundamental period*, the smallest value of N_0 that satisfies

$$x[n] = x[n + N_0].$$

Similar to continuous-time power and energy, the absolute value function within the sum is only necessary when the signal is complex-valued.

3.2 Useful Signal Operations

Standard signal operations like time-shifting, time-reversal are similar to those in continuous-time signals:

$$x_s[n] = x[n - 5]$$
$$x_r[n] = x[-n].$$

The discrete analogue of compression and expansion of continuous-time signals can be referred to down-sampling and up-sampling. Since discrete-time signals have a discrete domain, compression and expansion result in *less* or *more* values of $x[n]$ when compared to the original signal. For example, down-sampling by a factor of 2 (compression), every other sample is removed; in up-sampling (expansion) by a factor of 2, intermediate samples would be added between each original value are inserted for a total of $2n$ samples when compared to the original signal. We could set the intermediate, added samples to 0, but it's common to artificially interpolate the average value of the two adjacent samples at each added sample (other, more accurate interpolation methods may be used as well). It's important to remember that when interpolating, we haven't added any *real* data. Likewise, it's important to note that in down-sampling, we will lose real data in the process.

3.3 Some Useful Discrete-Time Signal Models

Similar to continuous time signals, there are a number of useful signal models that correspond to their continuous-time counterparts.

The discrete-time version of the impulse function $\delta[n]$ is defined as

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Note that there is no infinitely-valued pulse for this version of the impulse function; instead, it's simply valued at 1 for $n = 0$, and zero elsewhere.

The unit step, similarly, is defined as

$$u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0. \end{cases}$$

Note that there is no ambiguity in the discrete-time case as to the value $u[0]$, as there is in continuous-time; instead, $u[0] = 1$.

The discrete-time exponential, analogous to the continuous-time exponential, $e^{\lambda t}$, requires some explanation. Though the fundamental value is the same, rather than converting directly to the discrete time domain as $e^{\lambda n}$, we will use γ^n , where $\gamma = e^\lambda$.

Discrete-time sinusoids are straightforward, and may be expressed as

$$C \cos(\Omega n + \theta).$$

Even though the resultant values will be discretized, the result will follow a sinusoidal pattern, and increased sampling rates will increase the accuracy of the result.

Extending to complex exponentials, Euler's formula still applies in the discrete case:

$$e^{j\Omega n} = \cos(\Omega n) + j \sin(\Omega n).$$

3.4 Examples of Discrete-Time Systems

As specific examples of discrete-time systems were covered in depth in Chapter 1, we won't revisit them here.

However, it is worth pointing out a concept analogous to the differential equation in continuous-time systems, the **difference equation**. A comparison between the two is better served by an example.

Given a simple, first-order differential equation

$$\frac{dy(t)}{dt} + cy(t) = x(t),$$

we will consider uniformly sampling $x(t)$ at intervals of T seconds, transforming it into $x[n]$. We'll do the same for $y(t) = y[n]$. Then, using the definition of the derivative with a discrete-time twist,

$$\lim_{T \rightarrow 0} \frac{y[n] - y[n-1]}{T} + cy[n] = x[n].$$

Simplifying,

$$y[n] + \alpha y[n-1] = \beta x[n],$$

where

$$\alpha = \frac{-1}{1 + cT} \quad \text{and} \quad \beta = \frac{T}{1 + cT}.$$

It will be convenient later on to use what we will refer to as the "advance" form of a difference equation, which can be accomplished by simply time shifting both sides of the equation such that $y[n]$ is the lowest order sample in the equation. For example, delaying the above example by 1,

$$y[n + 1] + \alpha y[n] = \beta x[n + 1].$$

As we'll see, this notation makes it easier to make certain assumptions about the memory requirements of the discrete-time system under consideration.

Similar to continuous-time signals, discrete-time signals can be categorized as linear/non-linear, time-variant/time-invariant, and so on.

3.5 Discrete-Time System Equations

Now that the groundwork is laid for discrete-time signals and their corresponding systems, we can move on to develop a procedure for finding a discrete-time system's ZIR and ZSR, where its total response will also be the sum of the two. As noted before, we will be working with difference equations rather than differential equations in this chapter, but we will still be sticking to *linear* difference equations with constant coefficients.

We have the choice of using the advance and delay forms of a difference equation, but we'll be working primarily with the advance form for convenience (the choice is somewhat arbitrary). For reference, the standard form of an N, M -order difference equation is

$$\begin{aligned} y[n + N] + a_1 y[n + N - 1] + \dots + a_{N-1} y[n + 1] + a_N y[n] = \\ b_0 x[n + N] + b_1 x[n + N - 1] + \dots + b_{N-1} x[n + 1] + b_N x[n]. \end{aligned}$$

As with the continuous-time discussion, $M = N$ at most, and more often than not, $M < N$; both ensure that the system remains causal. Generally, we will normalize the coefficient of the highest order term, a_0 , to 1, by dividing it and all the other coefficients by the appropriate value. This ensures that we can safely time-shift terms, as in the initial example, without causing problems.

To solve a difference equation, we can resort to an iterative or recursive method. Similar to a continuously-valued differential equation, the solution to $y[0]$ requires knowledge of some initial conditions; that is, the system's prior outputs (if we use the delay form of the difference equation) $y[n - 1], y[n - 2], \dots, y[-N]$, and the right hand side can be solved similarly. Therefore, a recursive or iterative method is often appropriate to solve a given difference equation, where previous values of the input/output are used to determine the corresponding value at $n = 0$. Then, each future step can also be solved for using the previous step, and so on, as long as we have N initial conditions available to us to solve for $y[0]$.

In continuous-time systems, we used the D operator to represent differentiation. Here, we'll use E as the **advance operator**, where

$$\begin{aligned} Ex[n] &= x[n + 1] \\ E^2 x[n] &= x[n + 2], \end{aligned}$$

and so on.

Then, we can represent any discrete system's difference equation as

$$Q[E]y[n] = P[E]x[n].$$

3.6 System Response to Internal Conditions: The Zero-Input Response

Similar to continuous-time differential equations, we'll handle the zero-input response by setting $x[n] = 0$,

$$Q[E]y_0[n] = 0.$$

Recall that earlier in the chapter, we defined $e^{\lambda n} = \gamma^n$, where $e^\lambda = \gamma$. Therefore, the solution to y_0 is now represented as

$$y_0[n] = c_1\gamma_1^n + c_2\gamma_2^n + \dots + c_n\gamma_N^n.$$

As in continuous time, the ZIR will be determined in total by the roots,

For example: consider the LTID system

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2].$$

Setting the input to $x[n] = 0$ and considering it to be *causal* (**note: why is this important?**), with initial conditions $y[-1] = 0$ and $y[-2] = \frac{25}{4}$, we will use E notation,

$$(E^2 - 0.6E - 0.16)y[n] = 5E^2x[n],$$

which yields roots

$$\gamma^2 - 0.6\gamma - 0.16 =$$

$$\gamma_1 = -0.2 \qquad \qquad \qquad \gamma_2 = 0.8.$$

Then, finding c_1 and c_2 using initial conditions,

$$c_1 = \frac{1}{5} \qquad \qquad \qquad c_2 = \frac{4}{5},$$

so that

$$y_0[n] = \frac{1}{5}(-0.2)^n + \frac{4}{5}(0.8)^n \quad n \geq 0.$$

Analysis of the characteristic roots $\gamma_1, \gamma_2, \dots, \gamma_n$ will yield information about the system's stability, but the criteria shifts from right- and left-half plane to whether the root is less than, equal to, or greater than unity (on the unit circle). (**note: look into rationale for this more**). As in continuous-time systems, pay close attention to whether the roots are real, repeated, and/or complex; this affects the form of your final solution.

The above methodology is different from the iterative method, since any n can be plugged into the equation. This is the analogue of the analytical solution to a differential equation, as opposed to the numerical solution.

3.7 The Unit Impulse Response $h[n]$

The impulse response for discrete-time systems is, analogous to continuous-time systems,

$$h[n] = \frac{b_N}{a_N}\delta[n] + y_c[n]u[n].$$

Comparing to continuous systems, $b_0 = \frac{b_N}{a_N}$ and $y_c[n]$ is a linear combination of the characteristic modes $\gamma_1, \gamma_2, \dots, \gamma_n$, which will depend on the system's characteristic modes. As before, we will not need $\frac{b_N}{a_N}$ if $M < N$, since it is 0.

How do we find the initial conditions $c_1, c_2, \dots, c_n$? Remember that before the impulse is applied, the system is in its zero state, but after the impulse is applied, new initial conditions are set. So, we will need two values of $h[n]$ to find these new conditions. We can use the iterative method to find these conditions, such that $h[-1] = h[-2] = 0$, then using these two values for $n = 0$ and $n = 1$ to find $h[0]$ and $h[1]$ and solve

for $c_1, c_2, \dots, c_n$ using the general solution. (**note: review this in much more detail, and go back and review impulse matching for continuous-time systems too, might make more sense now with the comparison available between the two types of system.**)

For example: find the impulse response $h[n]$ for $y[n+1] - y[n] = x[n]$. We know that the general form of $h[n]$ is $\frac{b_N}{a_N} \delta[n] + y_c[n]u[n]$. Here, $a_N = -1$ and $b_N = 1$, so $\frac{b_N}{a_N} = -1$. Now, we know $h[n] = -\delta[n] + y_c[n]u[n]$. Now, we find $y_c[n]$, the linear combination of the characteristic modes γ_1 and γ_2 . Using the characteristic equation,

$$\begin{aligned}(E - 1)y[n] &= x[n] \\ \gamma - 1 &= 0 \\ \gamma_1 &= 1.\end{aligned}$$

note: why only one mode? Using this mode, we now have $y_c[n] = k\gamma^n = k \times 1^n = k$, since there is only one characteristic mode. Now, we need to find k . Using the iterative method mentioned earlier,

$$\begin{aligned}y[n] - y[n-1] &= x[n-1] \\ h[n] - h[n-1] &= \delta[n-1] \\ h[0] - h[-1] &= \delta[-1] \quad h[0] = 0.\end{aligned}$$

Therefore, we can plug in $h[0] = 0$ now, and

$$h[0] = -\delta[0] + ku[0] \Rightarrow -1 + 0 = k \Rightarrow k = 1.$$

Then we can find the impulse response:

$$h[n] = -\delta[n] + u[n].$$

Why doesn't this match the textbook answer?

If you draw out the $h[n]$ that we found, you'll see that it reduces to $h[n] = u[n-1]$, so both are correct, just formulated differently.

Note: Apparently we call them natural modes not characteristic modes in discrete time. Nevermind not sure about this now.

Review of closed-form determination of the impulse response (see Drill 3.14 in textbook).

3.8 System Response to External Input: The Zero-State Response

To find the ZSR of the system given an input signal $x[n]$, we use the same property as a continuous-time system, by convoluting the input signal with the impulse response,

$$x[n] * h[n] = y[n].$$

Review differences in the width property for discrete vs continuous time.

Discrete-time convolution follows similar steps to continuous time convolution. First, you'll "flip" one function, $g[m] = g[-m]$ (this mirrors the function about the y axis. We can begin to shift $g[-m]$, and there can be no overlap, partial overlap, and complete overlap between the two signals. Rather than evaluating the function, though, we will sum at each discrete step of the shift.

This results in an iterative kind of solution. Can we find a closed-form solution? The general solution can be found in Chapter B of the textbook.

3.9 System Stability

As usual, the total response is ZIR + ZSR. Mathematically,

$$\sum_{j=1}^N c_j \gamma_j^n + x[n] * h[n].$$

Similarly, again, to the continuous-time case, BIBO stability can be determined by evaluating the impulse response of the system; is it absolutely summable? Mathematically, BIBO criteria is

$$\sum_{n=-\infty}^{\infty} |h[n]| < K_2 < \infty.$$

Again, internal stability can be used to make conclusions about BIBO (external) stability.

Internal stability itself depends on the ZIR of the system (its characteristic roots/modes.)

Quick review of complex math, polar v rectangular form. As a consequence of how polar complex numbers work, we can assume that stability criteria for discrete time is

$$\begin{aligned} |\gamma| &< 1 \\ |\gamma| &> 1 \\ |\gamma| &= 1, \end{aligned}$$

where each case determines stable, marginally unstable, and unstable systems.

In short, you may determine internal stability using the values of γ that are found by solving the system's characteristic equation.

3.10 Exam review

Only first part of chapter 3, otherwise just Chapter 1, 2; lectures 1-7.

Chapter 1, signal classification, energy, power, etc. Equation sheet will be provided. No graphical convolution, thank fucking god. Signal energy and power, what is an energy signal vs power signal. Signal operations, shifting, scaling, etc. Classification of signals (easier than systems), using the unit step to describe piecewise functions (this will **definitely** be on the test, and we will need to know how to do slopes, etc. Probably good to review slope-intercept and point-slope form). System classification, we may need to show math for is it time-variant, etc. for causal, memoryless, maybe just need to write out justification, no math. total response is ZIR + ZSR.

Chapter 2, we will need to find system response given differential equation. Impulse matching (simplified) equation is given. Just know how to use them, find zero-input response and impulse response. Use response to find system stability.

From Chapter 3, we will need to know the following: Power and energy for discrete-time signals, impulse and unit functions in discrete time, he will either give us an equation and ask for the graph, or give us the graph and ask for the equation (unit step in discrete time).

What problems should you focus on more?

1. Represent a CT signal using $u(t)$.
2. System classification (linearity (use math justification), causality, time variance, memoryless, causal, analog/digital, invertible, stability).

3. Internal and external stability analysis. Given a differential equation, tell me what the internal and external stability is, mathematically. Given the characteristic modes, not the impulse response. So we can assume that an internally stable/unstable system is externally stable/unstable.
4. Finding the impulse response of a system, given the differential equation.
5. DT signal representation (only one Chapter 3 question).
6. a little extra credit, as a treat.